

47 New Bicycles — Factory Mechanical State

Pre-use diagnostic framework (PDI)

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Abstract

Diagnosis of 47 new bicycles (pre-use) across five factory mechanical failure vectors: lubrication, wheel integrity, hydraulic compliance, assembly torque and drivetrain losses, with impact quantified in watts and braking distance.

Sample — the 5 failure vectors

Vector	Parameter	Factory	Optimal	Impact
V1 Lubric. cadena	P. pérdida 250 W	6-10 W	3,4-4,0 W	-2,6 a -6,6 W
V1 Lubric. rodam.	mu cinético	0,05-0,10	0,01-0,03	fricción 3-5x menor
V2 Tensión radios	Var. adyacentes	±20-30%	< ±10%	fatiga acelerada
V3 Complianc. hidr.	Dist. parada	+25%	ref.	+1,75 m a 30 km/h
V4 Torque	% fuera rango	~40%	0%	micro-fisura carbono
V5 Transmisión	P. acumulada	7,5-13,5 W	3,9-5,5 W	Δ 3,6-8,0 W

Dataset variables

power loss (W); friction coefficient; spoke tension variation (%); stopping distance; % components out of torque. The per-unit dataset (47 records with raw readings) is shared under agreement.

Limitations

The sample (47 units) is illustrative of mid/high-end new bicycles; it quantifies real vectors, not population validity. Open to expansion.

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Provisional by design. A verifiable sample is published; the full raw set and the reproducible method are shared under agreement. Contact: carlos.ravello@zoovettravel.com